



Major Project: Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.

However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



PROJECT DESCRIPTION

This project analyzes Seasonal Agriculture Performance to understand how agricultural outcomes change across different seasons. It studies the influence of environmental conditions, farming practices, resource usage, crop selection, and economic factors on agricultural performance.

Key Areas of Analysis

- **Data Cleaning:** This handles missing values, duplicates, data types and outliers.
- **Seasonal Analysis:** Here comparison of yield, production, revenue, cost, and profit across seasons takes place.
- **Relationship Analysis:** Here effect of rainfall, temperature, soil conditions, fertilizer, irrigation, and seed quality on crop yield is observed and analyzed.
- **Resource Analysis:** This handles water usage and water-use efficiency across irrigation methods and seasons.
- **Economic Analysis:** Here crop-wise and seasonal profitability and profit margins is analyzed.
- **Statistical & Correlation Analysis:** Identification of significant relationships and patterns takes place.
- **Visualizations:** Bar charts, box plots, scatter plots, heatmaps, correlation matrices, etc are used.

This project identifies and analyses seasonal trends, important agricultural factors, unusual patterns, and profitability differences leading to data-driven recommendations for better crop planning, irrigation, resource utilization, and seasonal agricultural decision-making and maximizing the profits.

WHO ARE THE END USERS?

This project can be useful for anyone involved in agricultural planning, monitoring and decision-making, such as :



- **Farmers** – They plan crops, irrigation, fertilizers and resources according to seasonal conditions.
- **Government & Agricultural Departments** – Support seasonal planning, policy making and resource allocation.
- **Agricultural Planners & Experts** – Analyze crop performance and identify trends.
- **Irrigation & Resource Managers** – Improve water utilization and optimize resource efficiency.
- **Researchers & Data Analysts** – Study relationships between environmental factors and agricultural outcomes such as yield, production, and profitability..
- **Agribusinesses & Agricultural Organizations** – Understand production, profitability and market-related patterns to support better decisions.

Overall, the projects is designed to support data-driven agriculture decision-making and better seasonal planning for maximizing profits and efficient resource utilization.



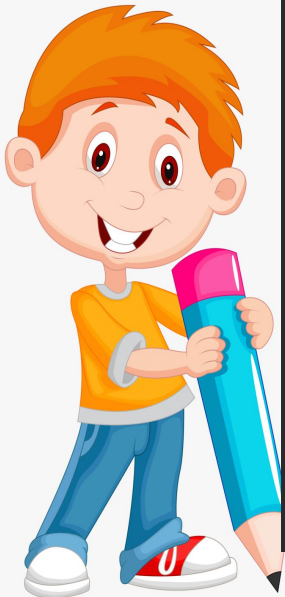
TECHNOLOGY USED

Core Environment & Libraries

- **Programming Language:** Python
- **Development Environment:** Google Colab
- **Data Processing & Manipulation:** Pandas, NumPy
- **Data Visualization:** Matplotlib, Seaborn
- **Statistical Analysis:** SciPy
- **Machine Learning:** Scikit-learn

Visualizations & Analysis Implemented

- **Bar & Count Plots:** Compared crop distribution, seasonal performance, irrigation methods, and regional variations.
- **Horizontal Bar Graphs:** Presented category-wise comparisons and rankings clearly.
- **Pie Charts:** Showed proportions and composition of categorical variables.
- **Histograms & Box Plots:** Analysed data distributions and identified outliers.
- **Scatter Plots:** Examined relationships between environmental, resource, and yield variables.
- **Pair Plots:** Visualized relationships and distributions among multiple numerical variables.
- **Heatmaps:** Displayed correlations between agricultural, environmental, and economic factors.
- **Raincloud/Violin-Box Hybrid Plots:** Compared data distribution, density, spread, and outliers across categories.
- **ECDF (Empirical Cumulative Distribution Function):** Analysed the cumulative distribution and probability of yield values.
- **Radar Charts:** Compared multiple performance metrics across different crops.
- **Correlation Analysis:** Identified relationships among environmental conditions, resource usage, yield, revenue, and profit.



TECHNOLOGY USED

Analytical Techniques Applied

- Data Cleaning & Missing Value Handling
- Duplicate & Outlier Detection
- Exploratory Data Analysis (EDA)
- Univariate, Bivariate & Multivariate Analysis
- Descriptive Statistics
- Correlation & Crop-wise Comparison
- Resource & Profitability Analysis
- Predictive Analysis using Machine Learning

Dataset

- **Source: Seasonal Agricultural Performance Dataset**
- The dataset represents agricultural activities carried out across different seasons, geographical areas and farming conditions. It contains information related to farming practices, environmental conditions, crop production, resource usage and economic performance. The dataset provides students with an opportunity to explore how agricultural performance changes across seasons and to identify meaningful patterns and differences through data analysis.
- The dataset contains farm-level agricultural records covering different seasons, locations, crops, environmental conditions, farming practices, production, costs, revenue, profit and resource usage.

Dataset file: `seasonal_agriculture_performance_dataset.csv`

- The size of dataset is such that it contains 4000 records with 20 variables
- The Key Features are Crop, Season, State, Rainfall, Temperature, Soil Conditions, Irrigation, Fertilizer, Yield, Production, Revenue, Profit, Water Usage, and Disease/Pest Risk.



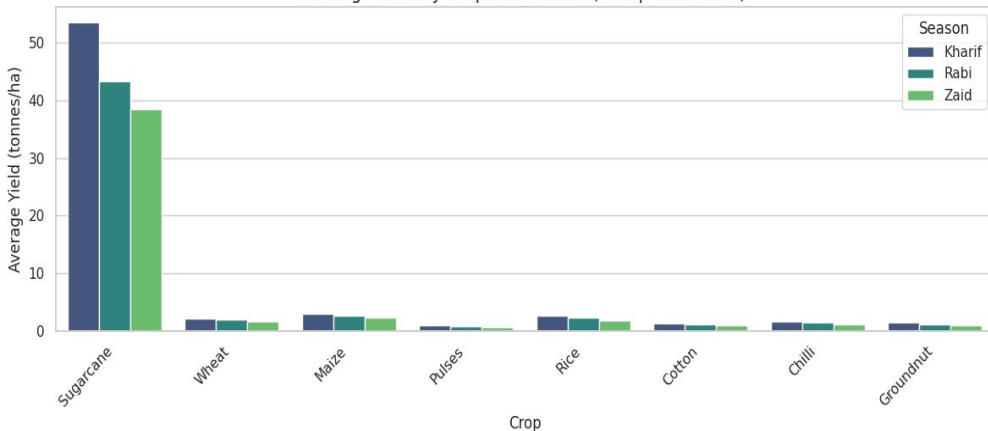
RESULTS: BAR GRAPHS

A bar graph is used to compare values of distinct categories with ease. The following graphs compare :

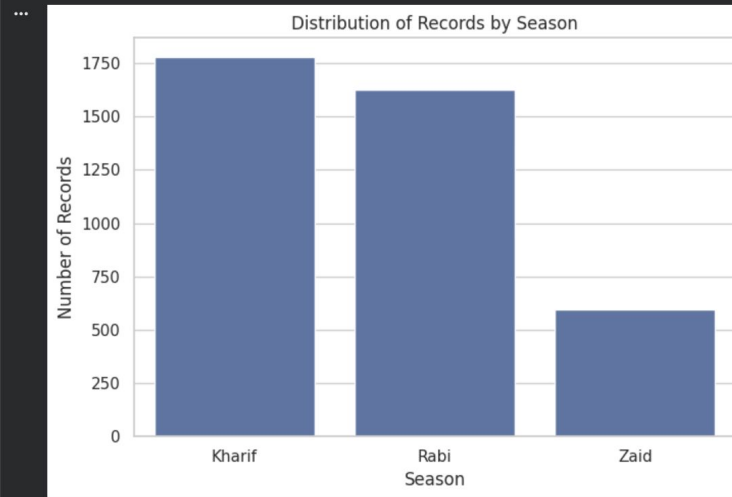
- The distribution of records by agriculture season.
- Highest and Lowest distribution of crops
- The average profit earned in each season
- Average Yield of Crop according to their respective season

Note: Bar graphs are used throughout the analysis and these are only a selection of various bar graphs.

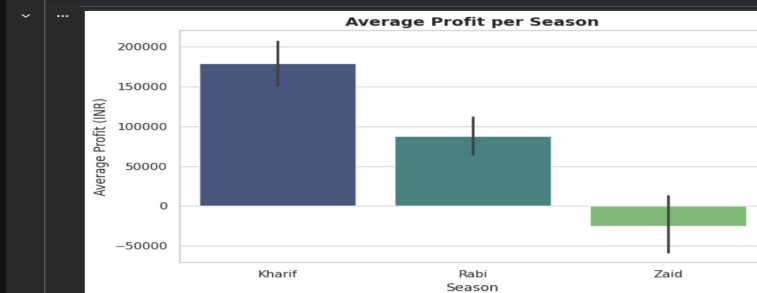
Average Yield by Crop and Season (Grouped Bar Plot)



```
# Univariate: Season distribution
plt.figure(figsize=(7, 5))
sns.countplot(data=df, x="Season")
plt.title("Distribution of Records by Season")
plt.xlabel("Season")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()
```



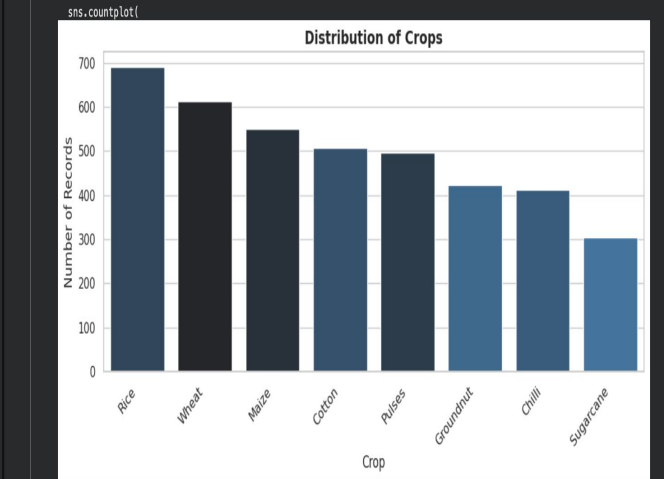
```
# Bivariate: Season vs Profit (Bar graph)
sns.set_theme(style="whitegrid")
plt.figure(figsize=(8, 5))
sns.barplot(
    data=df,
    x="Season",
    y="Profit_INR",
    estimator=np.mean,
    palette="viridis",
    hue="Season",
    legend=False
)
plt.title("Average Profit per Season", fontsize=14, fontweight="bold")
plt.xlabel("Season", fontsize=12)
plt.ylabel("Average Profit (INR)", fontsize=12)
plt.tight_layout()
plt.show()
```



```
# Univariate: Crop distribution
sns.set_theme(style="whitegrid")

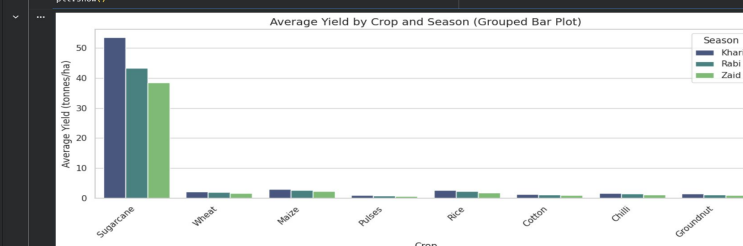
plt.figure(figsize=(10, 5))
sns.countplot(
    data=df,
    x="Crop",
    order=df["Crop"].value_counts().index,
    color="#1f77b4", # Professional deep blue (or use palette="mako" / palette="Blues_r")
    hue="Crop",
    legend=False
)
plt.xticks(rotation=45, ha="right")
plt.title("Distribution of Crops", fontsize=14, fontweight="bold")
plt.xlabel("Crop", fontsize=12)
plt.ylabel("Number of Records", fontsize=12)
plt.tight_layout()
plt.show()
```

Setting a gradient palette using color= is deprecated and will be removed in v0.14.0. Set 'palette="dark:#1f77b4"' for the same effect.



```
# Multivariate: Average Yield by Crop and Season (Grouped Bar Plot)
sns.set_theme(style="whitegrid")
# Sort crops so Sugarcane appears first on the x-axis
crop_order = df["Crop"].unique().tolist()
if "Sugarcane" in crop_order:
    crop_order.remove("Sugarcane")
    crop_order.insert(0, "Sugarcane")

plt.figure(figsize=(12, 5))
sns.barplot(
    data=df,
    x="Crop",
    y="Yield_Tonnes_Ha",
    hue="Season",
    order=crop_order,
    palette="viridis",
    errorbar=None
)
plt.title("Average Yield by Crop and Season (Grouped Bar Plot)", fontsize=14)
plt.xlabel("Crop", fontsize=12)
plt.ylabel("Average Yield (tonnes/ha)", fontsize=12)
plt.xticks(rotation=45, ha="right")
plt.legend(title="Season", loc="upper right")
plt.tight_layout()
plt.show()
```

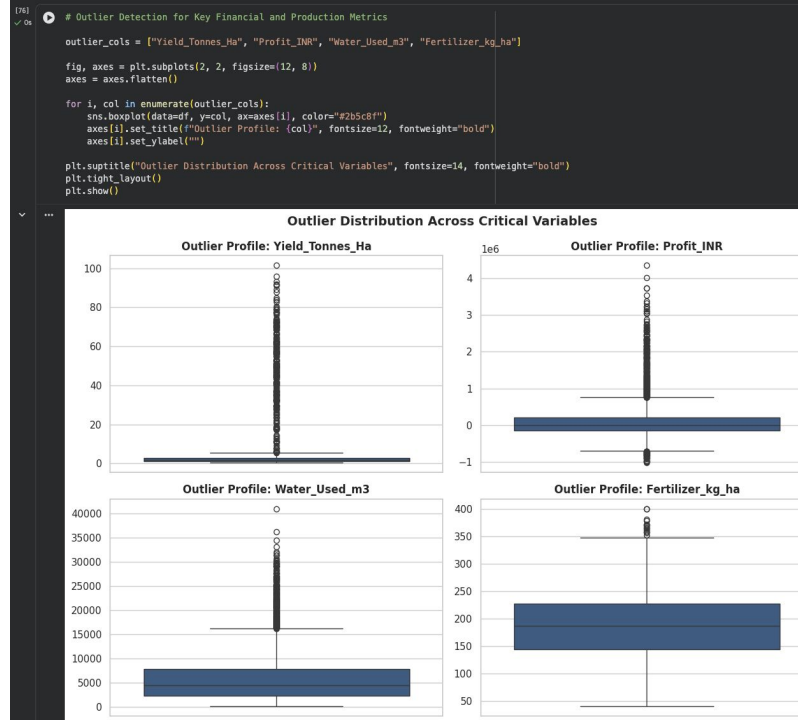
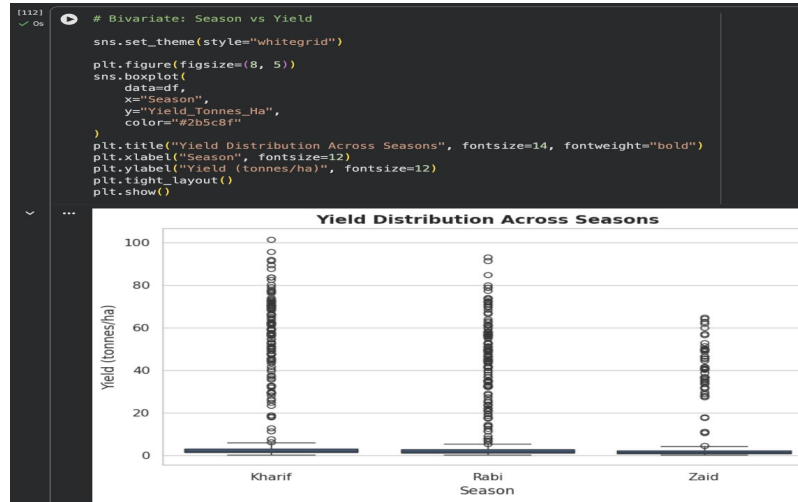


RESULTS- BOX PLOT

A box plot is used to understand the distribution, variability, median, and outliers in numerical data. The following graphs are used for :-

- Comparing yield across different agricultural seasons
- To identify & compare the fertilizer efficiency across different irrigation methods
- Identification of Outliers (unusual values in key variables i.e. yield, profit, water and fertilizer usage)
- To compare yield variation among different crops across seasons.

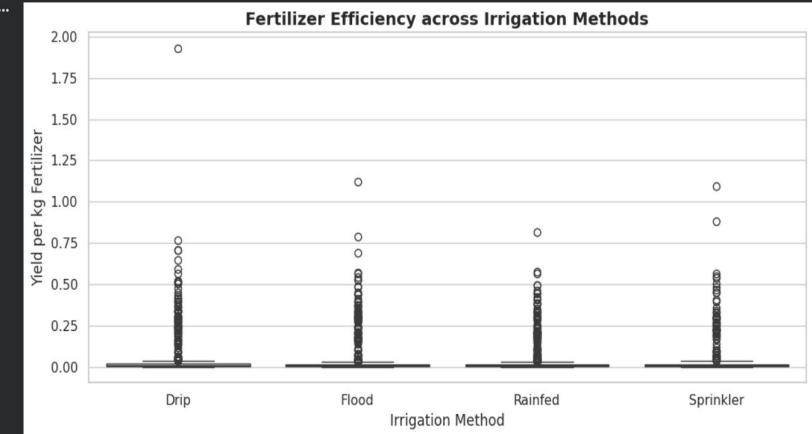
Note: Box plots are used throughout the analysis; the graphs shown here only represent a few of the many box plots generated.



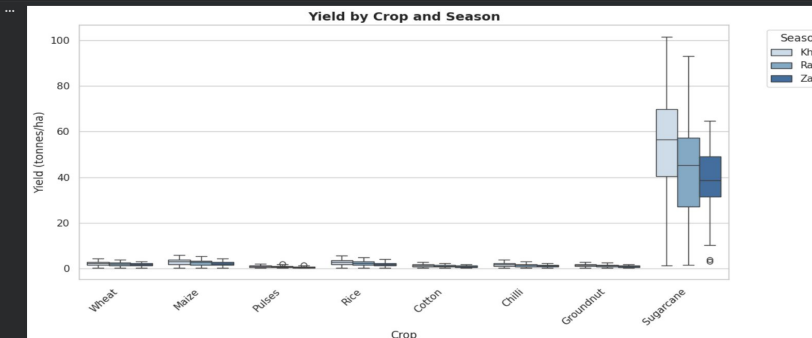
```
[183] ✓ Os # Student Analysis 2
# Question: What is the Resource ROI (Yield per unit of Fertilizer) across Irrigation Methods?
# Analysis: Evaluating resource use efficiency to minimize input costs.
# Visualization: Boxplot comparing fertilizer efficiency grouped by irrigation type.
# Interpretation: Shows which irrigation systems best convert fertilizer into crop yield.

if "Fertilizer_Efficiency" not in df.columns:
    df["Fertilizer_Efficiency"] = df["Yield_Tonnes_Ha"] / df["Fertilizer_kg_ha"]

plt.figure(figsize=(10, 5))
sns.boxplot(
    data=df,
    x="Irrigation_Method",
    y="Fertilizer_Efficiency",
    hue="Irrigation_Method", # Explicitly set hue to x-variable
    palette="Blues",
    legend=False # Suppress legend as it's redundant when hue=x
)
plt.title("Fertilizer Efficiency across Irrigation Methods", fontsize=14, fontweight="bold")
plt.xlabel("Irrigation Method", fontsize=12)
plt.ylabel("Yield per kg Fertilizer", fontsize=12)
plt.tight_layout()
plt.show()
```



```
[114] ✓ Os # Multivariate: Season, crop and yield
sns.set_theme(style="whitegrid")
plt.figure(figsize=(12, 6))
sns.boxplot(
    data=df,
    x="Crop",
    y="Yield_Tonnes_Ha",
    hue="Season",
    palette="Blues"
)
plt.xticks(rotation=45, ha="right")
plt.title("Yield by Crop and Season", fontsize=14, fontweight="bold")
plt.xlabel("Crop", fontsize=12)
plt.ylabel("Yield (tonnes/ha)", fontsize=12)
plt.legend(title="Season", box_to_anchor=(1.05, 1), loc="upper left")
plt.tight_layout()
plt.show()
```



RESULTS: PIE CHART

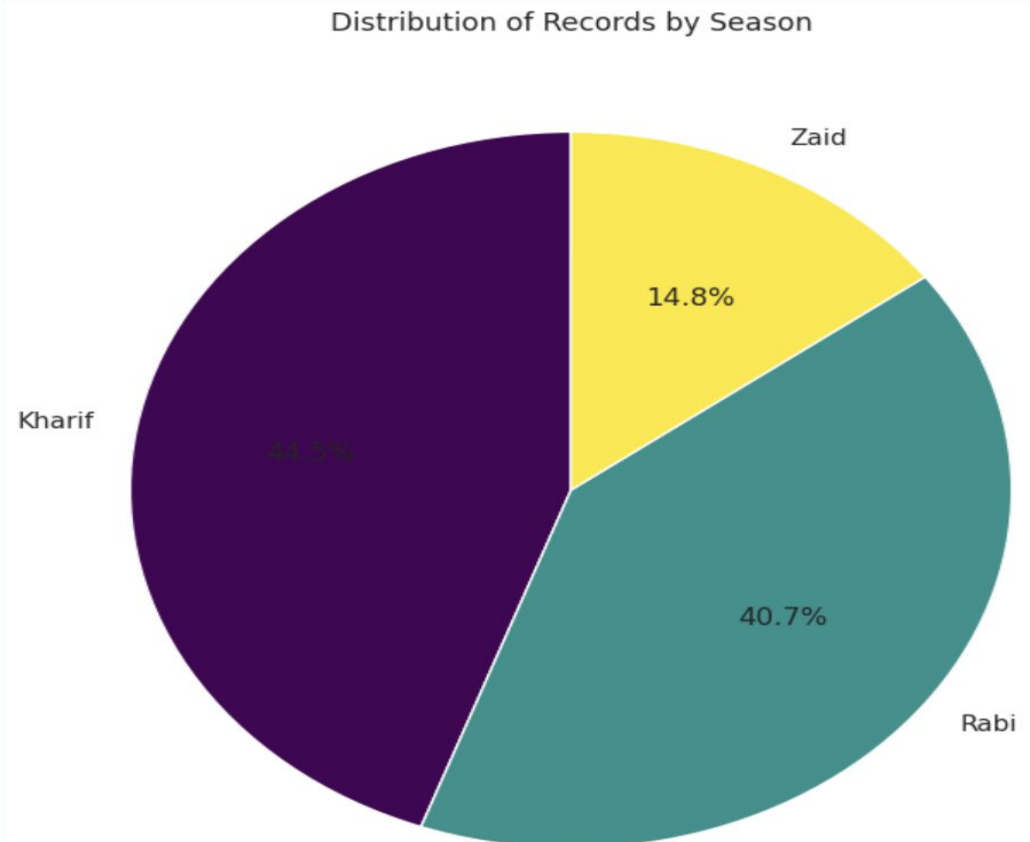
Pie chart is used to show how a whole is divided among different categories.

- The following pie chart shows the distribution of records which are spread out across the three agricultural seasons i.e. Kharif, Rabi and Zaid.
- The Kharif season has the largest portion of the records with 44.5% then comes Rabi with 40.7%. whereas Zaid has the smallest portion with 14.8% of all the records.
- **Key Insight:** Overall, the distribution Kharif and Rabi constitute the majority portion of the records and Zaid has a comparatively smaller representation.

Univariate: Season distribution (Pie Chart)

[74]
✓ 0s

```
plt.figure(figsize=(7, 7))
df['Season'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90, cmap='viridis')
plt.title('Distribution of Records by Season')
plt.ylabel('') # Hide the default 'Season' label on the y-axis
plt.tight_layout()
plt.show()
```



RESULTS-HISTOGRAM

Histogram are used to understand the distribution, frequency, and overall patterns of numerical data. The following graphs showcase:

- **Rainfall distribution** across different ranges, with most observations concentrated around moderate rainfall levels.
- Shows the most **agriculture yield** are concentrated at the lower end with only a few high values.
- Shows that the **water-efficiency** values are mainly concentrated near lower range
- Shows **profit per hectare** is also concentrated at lower values with a small number of high-profit observations.

Note: Histograms are used throughout the analysis; the graphs shown here only represent a few of the many histograms generated.

```
[62] # Distribution of Engineered Metrics (Water Efficiency & Profit Per Hectare)

# Calculate engineered metrics if not already present
if "Water_Efficiency" not in df.columns:
    df["Water_Efficiency"] = df["Production_Tonnes"] / df["Water_Used_m3"]

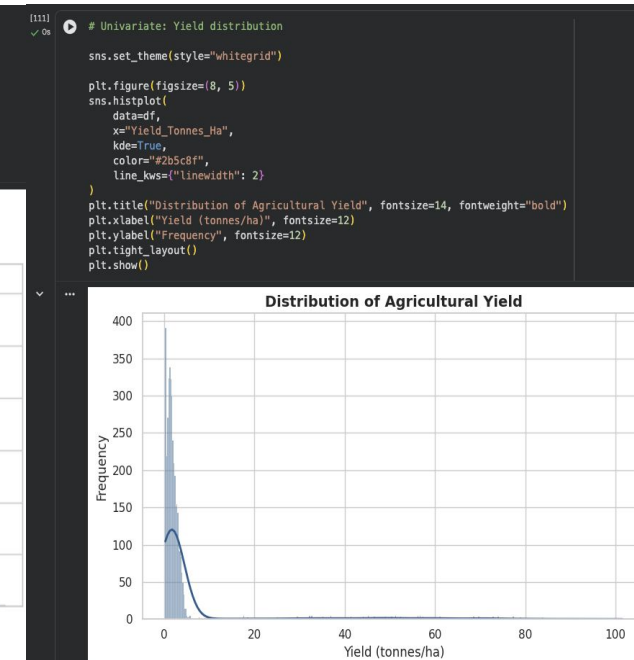
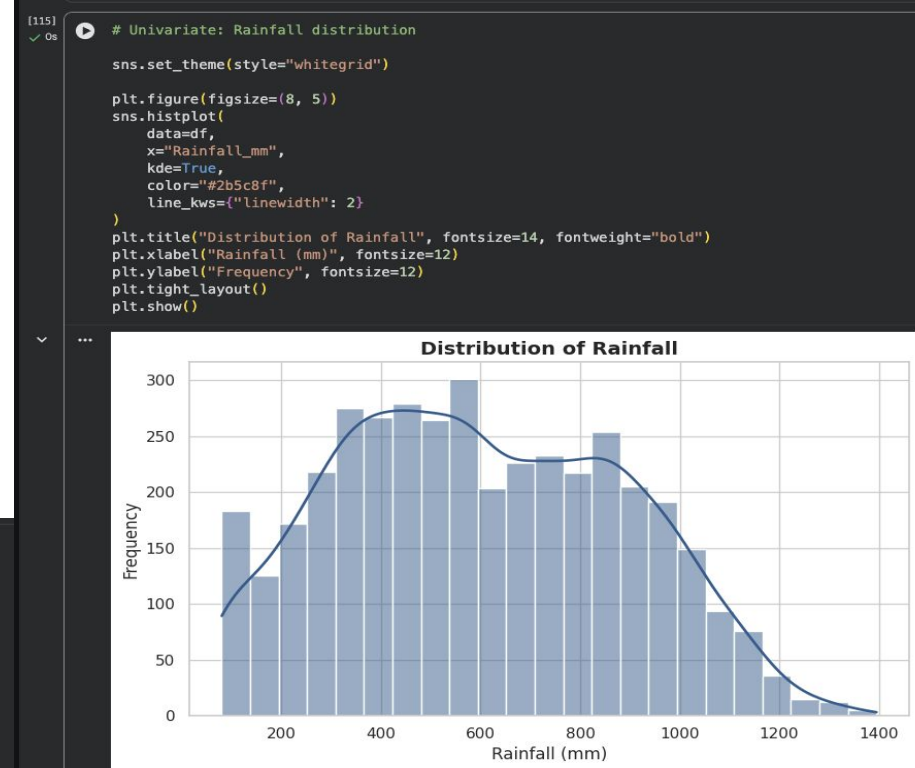
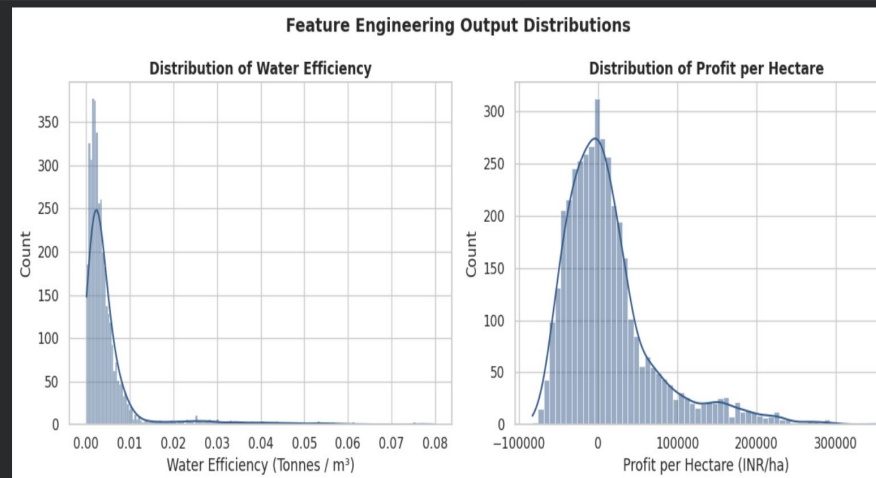
if "Profit_Per_Ha" not in df.columns:
    df["Profit_Per_Ha"] = df["Profit_INR"] / df["Farm_Area_Hectares"]

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.histplot(df["Water_Efficiency"], kde=True, ax=axes[0], color="#2b5c8f")
axes[0].set_title("Distribution of Water Efficiency", fontsize=12, fontweight="bold")
axes[0].set_xlabel("Water Efficiency (Tonnes / m³)")

sns.histplot(df["Profit_Per_Ha"], kde=True, ax=axes[1], color="#2b5c8f")
axes[1].set_title("Distribution of Profit per Hectare", fontsize=12, fontweight="bold")
axes[1].set_xlabel("Profit per Hectare (INR/ha)")

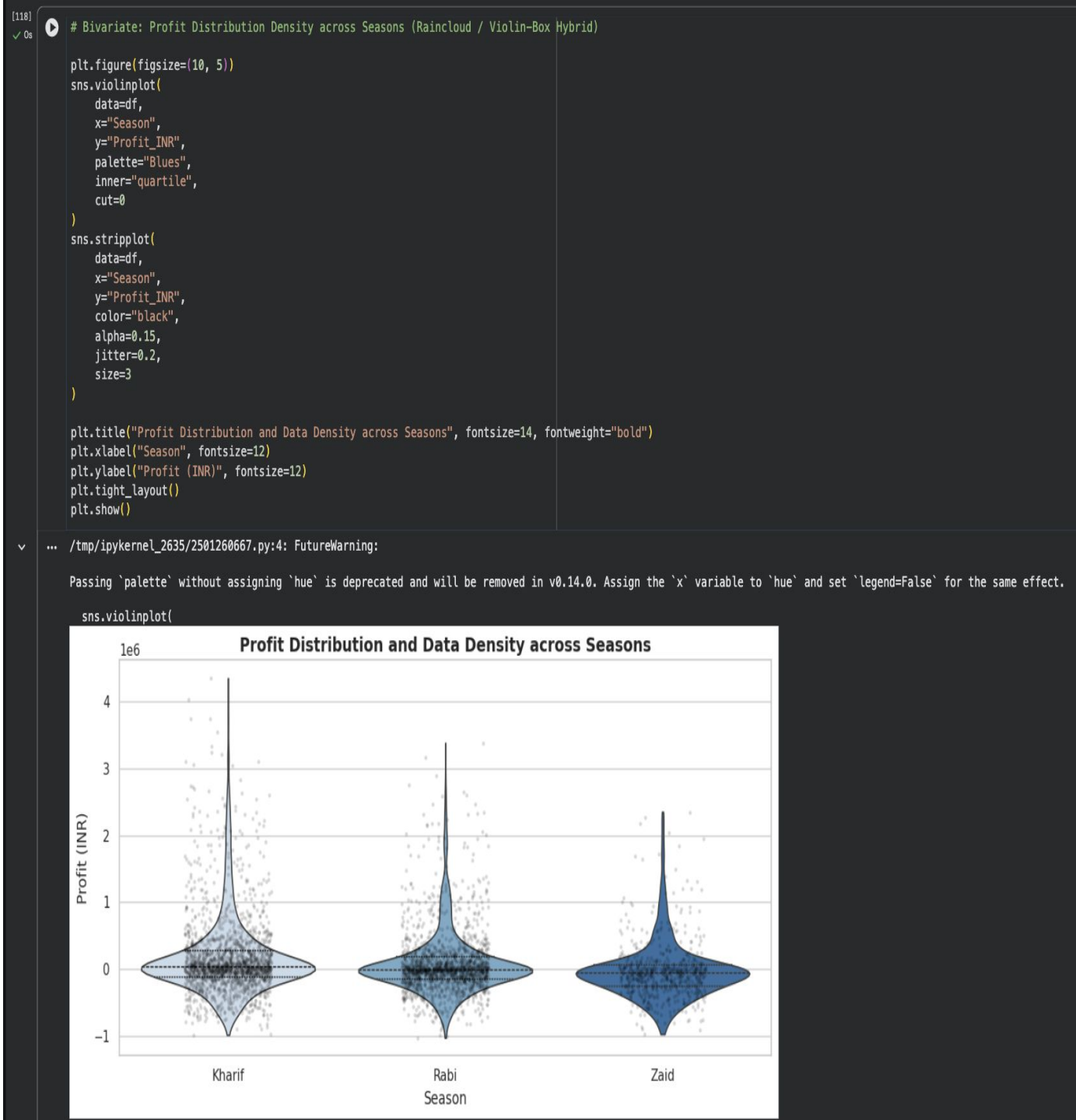
plt.suptitle("Feature Engineering Output Distributions", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.show()
```



RESULTS- RAINCLOUD/VIOLIN - BOX HYBRID PLOT

Rain-Cloud/Violin-Box Hybrid plot is used to show the distribution, density, spread, and outliers of numerical data across different groups. Here the graph shows the following:

- Compares profit distribution for Kharif, Rabi and Zaid agricultural seasons.
- The key insight is that Kharif shows the widest profit distribution, while Zaid has a comparatively lower range.
- Overall the plot provides a detailed view on both the central tendency and variation in profits across seasons.

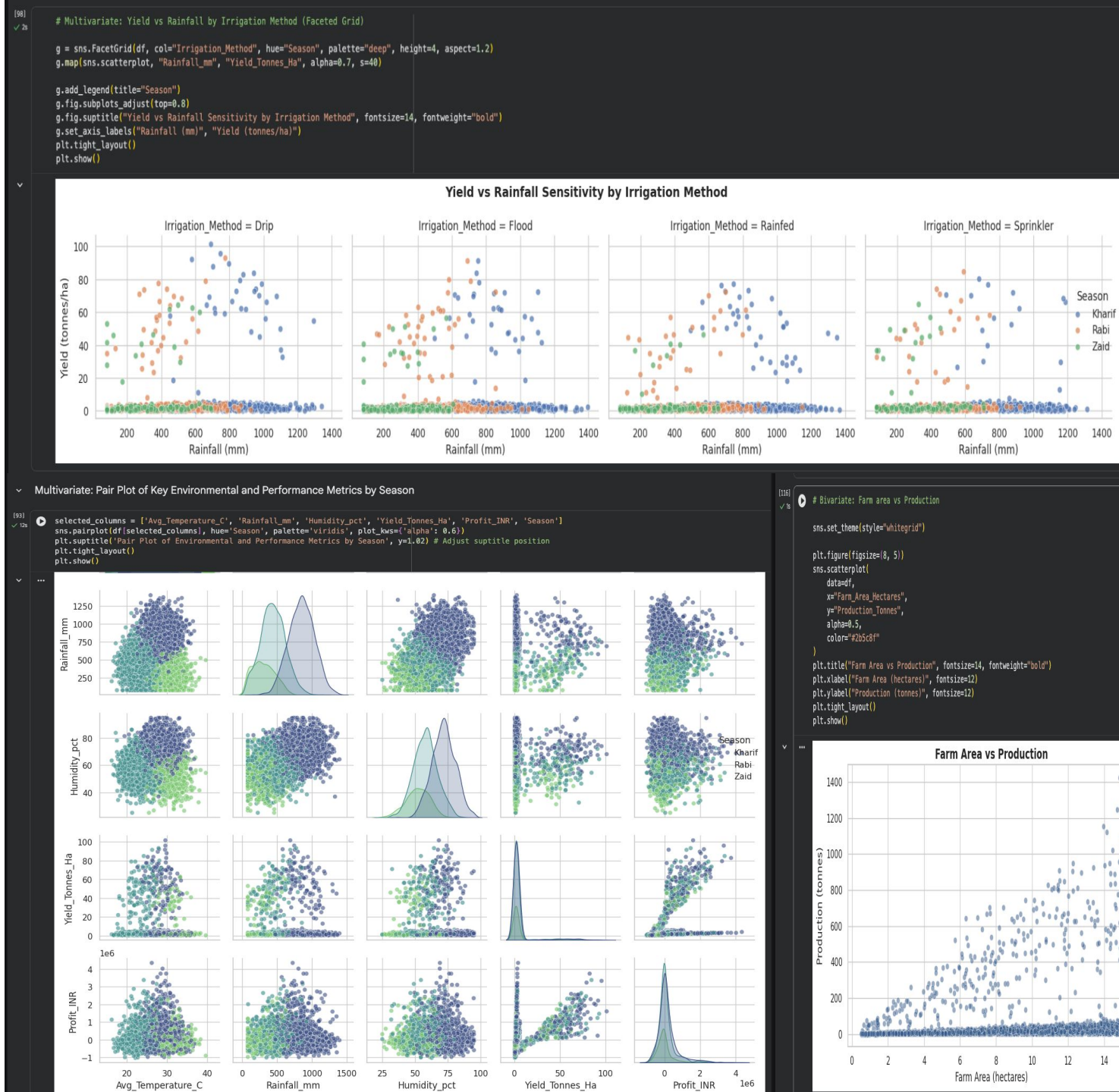


RESULTS-SCATTER PLOT & PAIR PLOT

Scatter plot is used to study the relationship between two numerical variables, while a pair plot shows relationships and distribution among multiple numerical variables at once. The following graphs here show that:

- It examines how rainfall relates to crop yield across different seasons and irrigation methods.
- It compares farm area and production and indicates a positive relationship and shows large farm areas generally correspond to higher production.
- Pair plot provides an overall view of relationships among rainfall, humidity, yield, profit, and other key variables, helping identify patterns and correlations.

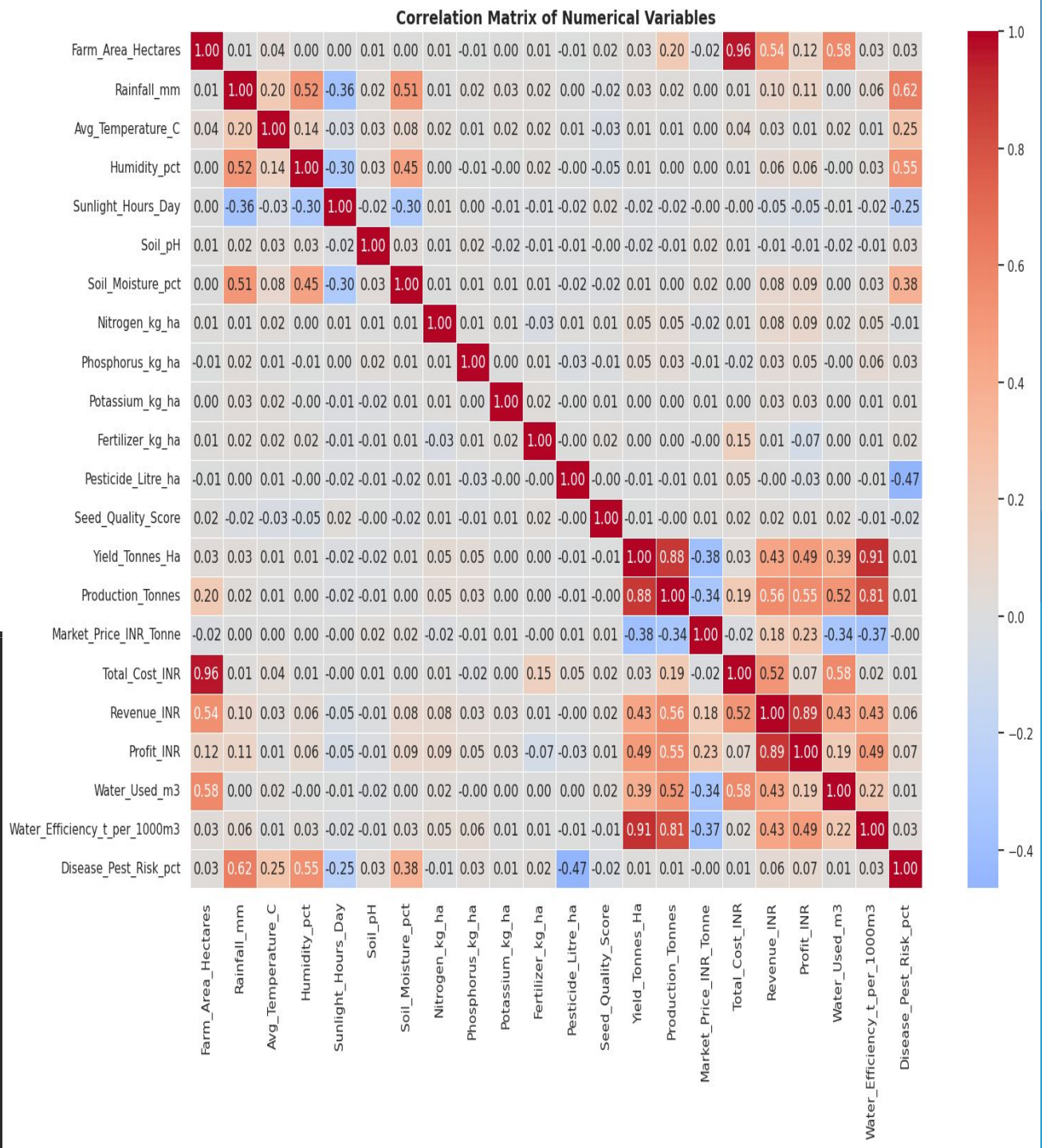
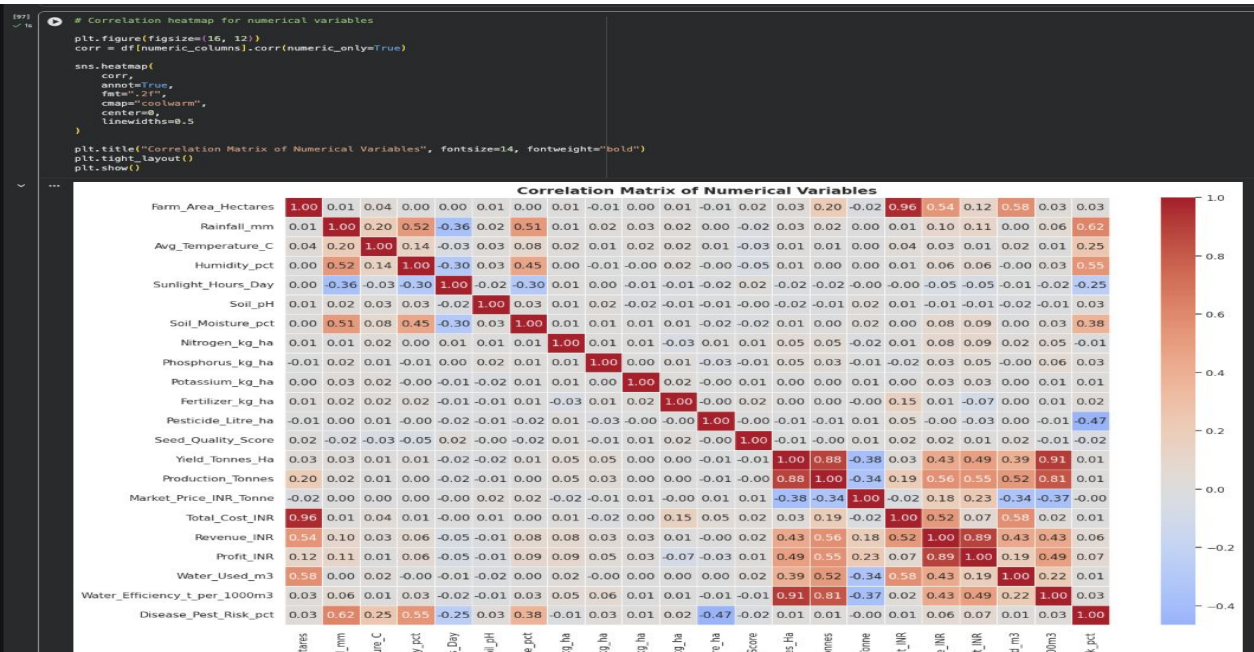
Note: Scatter plots and pair plots are used at various stages of the analysis; the visuals shown here represent only a few examples from the overall analysis.



RESULTS-HEATMAP

Heatmap is used to visualize the strength and direction of correlations between numerical values. Here the graph represents:

- Correlation matrix which highlights relationships among rainfall, yield, production, revenue, profit and other variables.
- Key insights which show the yield and production, as well as revenue and profit, also showcasing strong positive relationships.



RESULTS- HORIZONTAL BAR GRAPHS

Horizontal bar graphs is used to compare values across categories clearly, especially when category names are longer. The graph here showcases the following:

- Compares the percentage of profitable and loss-making farms for each crop.
- Cultivation records by crop shows the number of records for each crop, with Rice having the highest representation.

Note: Horizontal bar graphs are used at different stages of the analysis; these are only a few selected examples.

```
[117] ✓ Os
# Student Analysis 5: Financial Profitability vs Loss Ratio by Crop
df['Financial_Status'] = df['Profit_INR'].apply(lambda x: 'Profitable' if x > 0 else 'Loss-Making')

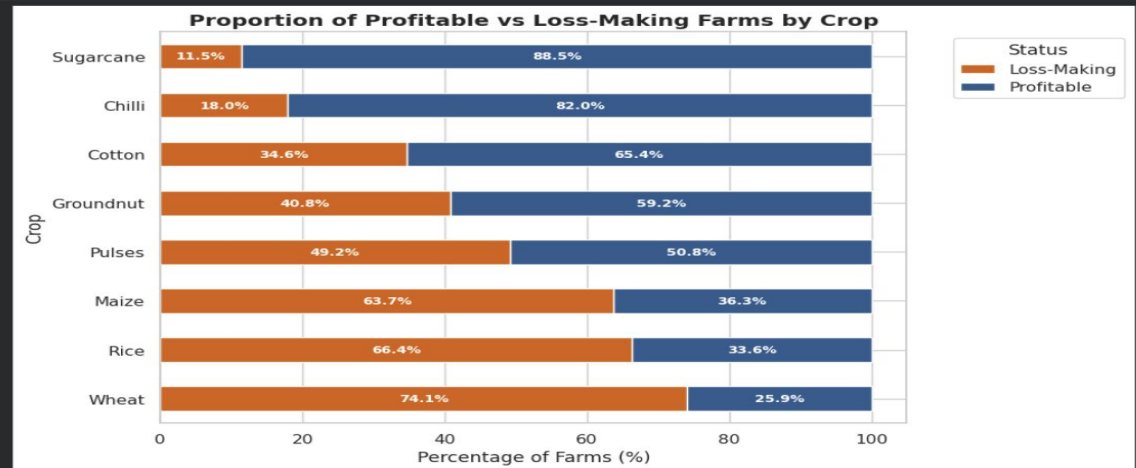
crop_status = pd.crosstab(df['Crop'], df['Financial_Status'], normalize='index') * 100
crop_status = crop_status.sort_values(by='Profitable', ascending=True)

ax = crop_status.plot(
    kind='barh',
    stacked=True,
    figsize=(10, 6),
    color=['#d95f02', '#2b5c8f']
)

plt.title("Proportion of Profitable vs Loss-Making Farms by Crop", fontsize=14, fontweight="bold")
plt.xlabel("Percentage of Farms (%)", fontsize=12)
plt.ylabel("Crop", fontsize=12)
plt.legend(title="Status", bbox_to_anchor=(1.05, 1), loc='upper left')

# Add percentage markers on the bars
for p in ax.patches:
    width, height = p.get_width(), p.get_height()
    if width > 5: # Only label if segment is wide enough
        x, y = p.get_xy()
        ax.text(x + width/2, y + height/2, f'{width:.1f}%', ha='center', va='center', color='white', fontweight='bold', fontsize=9)

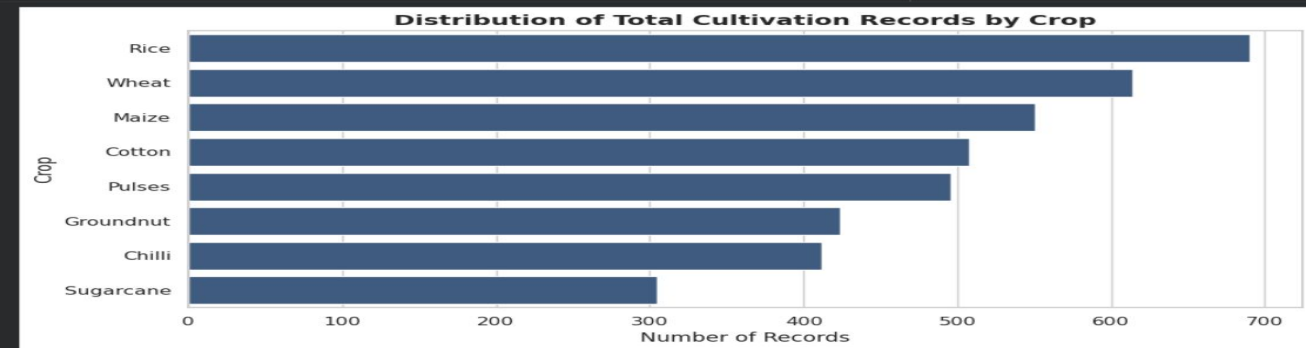
plt.tight_layout()
plt.show()
```



```
[73] ✓ Os
# Crop Cultivation Record Distribution (Univariate Count Plot)

plt.figure(figsize=(10, 5))
crop_order = df["Crop"].value_counts().index

sns.countplot(data=df, y="Crop", order=crop_order, color="#2b5c8f")
plt.title("Distribution of Total Cultivation Records by Crop", fontsize=14, fontweight="bold")
plt.xlabel("Number of Records", fontsize=12)
plt.ylabel("Crop", fontsize=12)
plt.tight_layout()
plt.show()
```



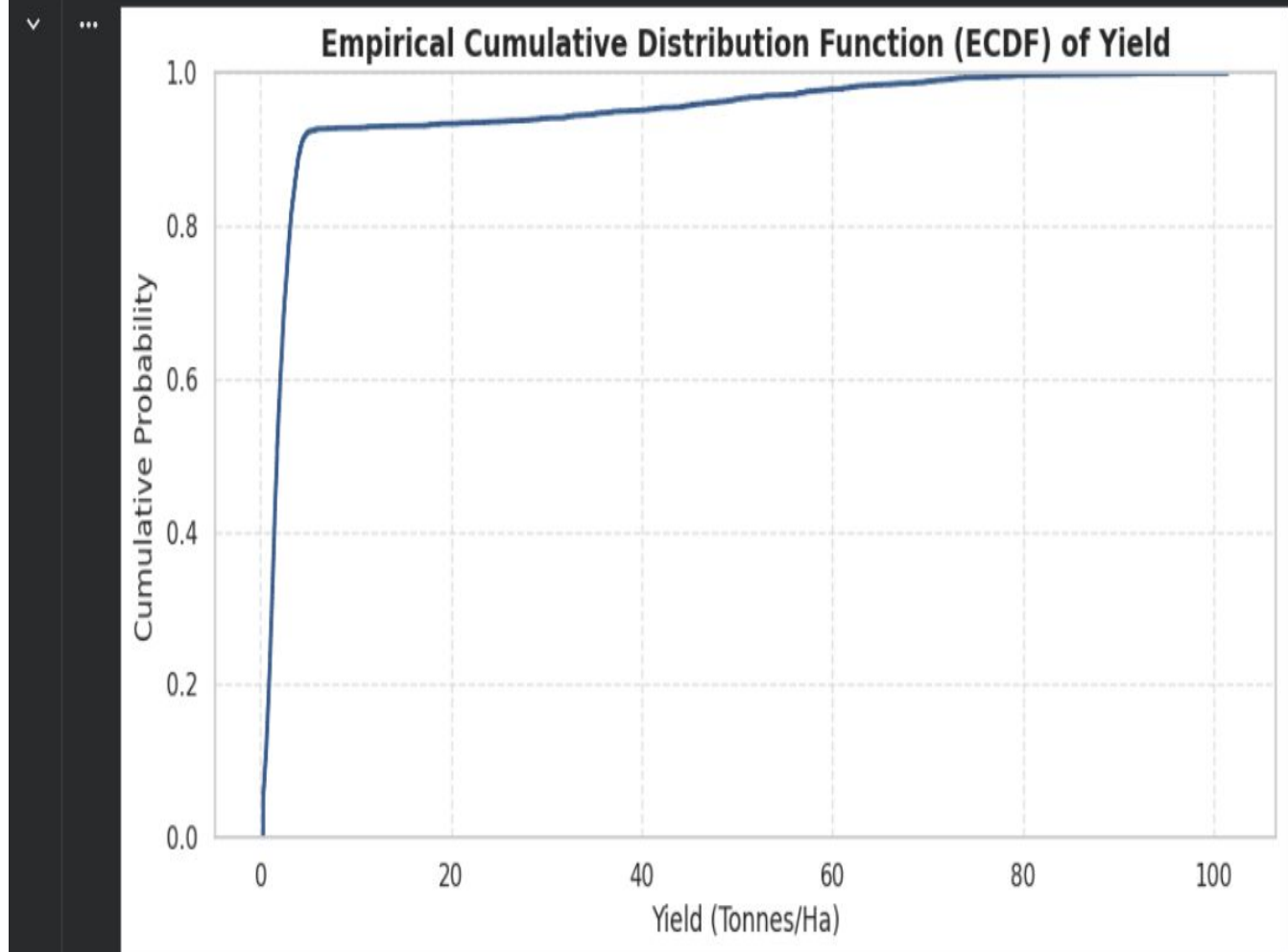
RESULTS- EMPIRICAL CUMULATIVE DISTRIBUTION FUNCTION(ECDF)

Empirical Cumulative Distribution Function(ECDF) is used to show the cumulative distribution of numerical data, helping understand what proportion of observations fall below a given value. Here the graph shows the following:

- Yield ECDF shows the cumulative proportion of farms at different yield levels.
- The key insight is that the curve rises sharply at lower yields, indicating the most observations have relatively low yield values.

```
[66] ✓ 0s # Empirical Cumulative Distribution Function (ECDF) for Yield

plt.figure(figsize=(9, 5))
sns.ecdfplot(data=df, x="Yield_Tonnes_Ha", color="#2b5c8f", linewidth=2)
plt.title("Empirical Cumulative Distribution Function (ECDF) of Yield", fontsize=14, fontweight="bold")
plt.xlabel("Yield (Tonnes/Ha)", fontsize=12)
plt.ylabel("Cumulative Probability", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.6)
plt.tight_layout()
plt.show()
```



RESULTS- RADAR CHART

Radar Chart is used to compare multiple performance measures across different categories in a single visual. The graph here showcases the following:

- The crop performance which compares different crops on the basis of yield, profitability, water efficiency, and fertilizer ROI.
- The key insight is that Sugarcane shows the strongest overall performance across the selected measures.

```
[185]
✓ 0s
# Student Analysis 4: Multi-Dimensional Performance Profiling across Key Crops (Radar Chart)
# 1. Select key performance metrics to compare across crops
radar_cols = ["Yield_Tonnes_Ha", "Profit_INR", "Water_Efficiency", "Fertilizer_Efficiency"]

# Ensure custom efficiency columns exist
if "Water_Efficiency" not in df.columns:
    df["Water_Efficiency"] = df["Production_Tonnes"] / df["Water_Used_m3"]
if "Fertilizer_Efficiency" not in df.columns:
    df["Fertilizer_Efficiency"] = df["Yield_Tonnes_Ha"] / df["Fertilizer_Used_kg"]

# 2. Aggregate average performance per crop
crop_profiles = df.groupby("Crop")[radar_cols].mean()

# 3. Min-Max normalize the metrics (scale 0 to 1) for fair multi-axis comparison
crop_profiles_norm = (crop_profiles - crop_profiles.min()) / (crop_profiles.max() - crop_profiles.min())

# 4. Select top representative crops to highlight on the radar plot
selected_crops = ["Sugarcane", "Chilli", "Rice", "Wheat"]
filtered_profiles = crop_profiles_norm.loc[selected_crops]

# 5. Radar Chart Setup
labels = ["Yield Volume", "Profitability", "Water Efficiency", "Fertilizer ROI"]
num_vars = len(labels)

angles = np.linspace(0, 2 * np.pi, num_vars, endpoint=False).tolist()
angles += angles[:1] # Complete the circle loop

fig, ax = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))

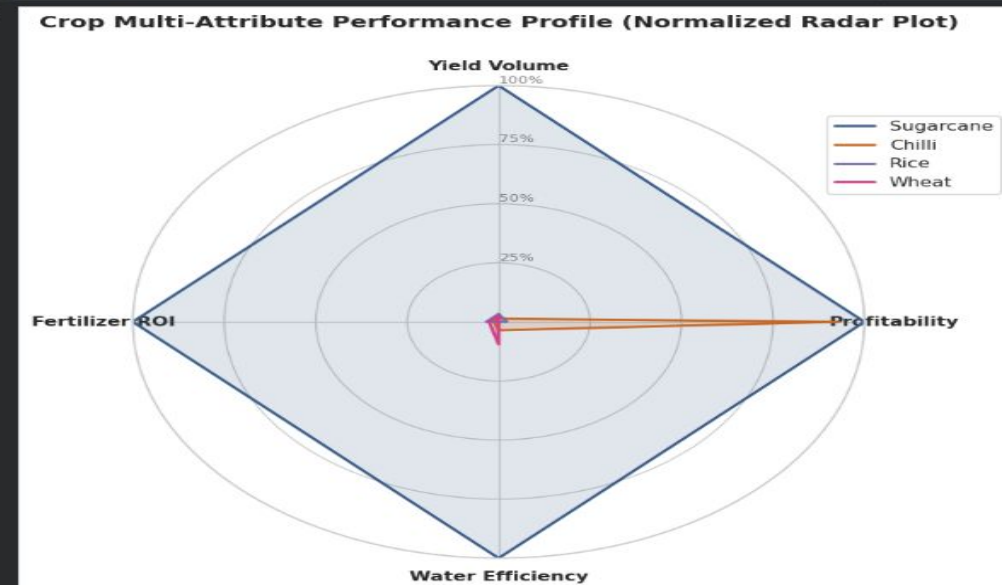
# Clean corporate palette mapping
colors = ["#2b5c8f", "#d95f02", "#7570b3", "#e7298a"]

for (crop_name, row), color in zip(filtered_profiles.iterrows(), colors):
    values = row.values.tolist()
    values += values[:1] # Complete the circle loop
    ax.plot(angles, values, linewidth=2, linestyle='solid', label=crop_name, color=color)
    ax.fill(angles, values, color=color, alpha=0.15)

# Formatting axes and style
ax.set_theta_offset(np.pi / 2)
ax.set_theta_direction(-1)
plt.xticks(angles[:-1], labels, size=11, fontweight="bold")
ax.set_rlabel_position(0)
plt.yticks([0.25, 0.50, 0.75, 1.0], ["25%", "50%", "75%", "100%"], color="grey", size=9)
plt.ylim(0, 1)

# Title shifted higher and Legend shifted slightly below the title line
plt.title("Crop Multi-Attribute Performance Profile (Normalized Radar Plot)", fontsize=14, fontweight="bold", pad=35)
plt.legend(loc="upper right", bbox_to_anchor=(1.20, 0.95))

plt.tight_layout()
plt.show()
```



FUTURE SCOPE

The **Seasonal Agriculture Performance Analysis** can be extended into a more advanced, real time agricultural decision-support system. The following are the future scope of this project :-

- **Predictive Modeling & Machine learning:** By implementing regression models (eg., Random Forest, XGBoost) we can forecast crop yield and profit on the basis of seasonal weather patterns, soil conditions, and resource inputs.
- **Prescriptive Analytics & Crop Recommendation System:** Developing a recommendation system which would suggest the most suitable crop based on soil type, pH, rainfall, temperature, water availability, and historical yield.
- **Smart Agriculture :** We can integrate AI, and IoT-based solutions ,live weather APIs for real time monitoring and data driven farming decisions to use rainfall, temperature, humidity, and weather forecasts for better seasonal analysis and prediction.
- **Accurate Farming & Input Optimization:** By building optimization algorithms to calculate precise fertilizer, pesticide, and water requirements which would minimizing agricultural runoff and maximizing return on investment while maintaining high yields.
- **Dynamic Market Pricing Integration:** We will integrate real-time market price data, subsidies, transportation costs, supply-demand trends and all the other costs to help farmers time their harvest sales for maximum revenue generation and reduce unnecessary losses.

GitHub Link

https://github.com/pragyasrivastavaaa/Major-Project-Seasonal-Agriculture-Performance-Analysis_VOIS-INTERNSHIP.git



VOIS Course completion certificate - Course name- Data Visualization



Vodafone Idea Foundation

VOIS

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foundation

Certificate of Completion

Presented to

Pragya Srivastava

For the successful completion of

Data Visualization

Issued on August 27, 2026

ID:VFLMS26_163323





THANK YOU